

Sample test (75 marks)

Name: _____ Date: _____

Answer all questions in the spaces provided.

1. Write the name of the state of matter represented by the pictures below.

A

B

C



(1.5 Marks)

2. Complete the table below.

	Draw the arrangement of particles	Describe the movement of particles	Does it change shape when poured?	Does it change volume when the size of the container changes?	Can it be compressed?
solid					
liquid					
gas					

(3 marks)

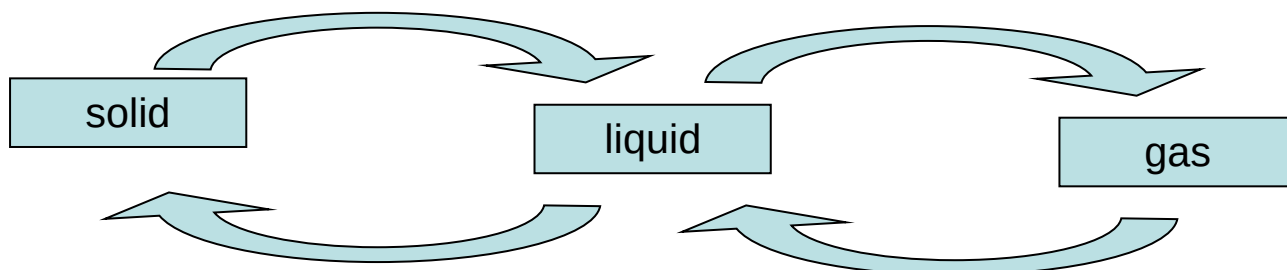
(3 marks)

(1.5 marks)

(1.5 marks)

(1.5 marks)

3. a) Complete the diagram below by writing for each arrow:
- what causes it to change state,** and
 - the name of the change.**



(8 Marks)

- b) Describe what is happening to the particles of matter when changing between each state.

(4 Marks)

4. Changes that occur in substances can be either physical or chemical. What are the differences between physical and chemical changes?

(2 Marks)

5. Write beside each letter the type of change (physical or chemical) pictured.

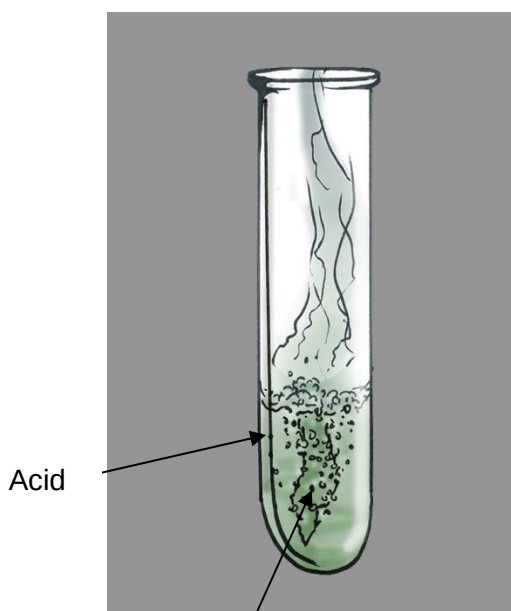
A.



B.



C.



D.



(4 marks)

6. If two substances are mixed together, how can you tell if a chemical change has taken place?

(5 Marks)

7. Write each of the following substances in the correct column of the table below.

air
sulfur
carbon dioxide

water
a cup of coffee
salt solution

iron sulfide
magnesium

zinc
soil

Element	Compound	Mixture

(10 Marks)

8. Complete the table for each compound listed below.

Formula	Number and name of atoms
HNO ₃	
HCl	
CuCO ₃	
NaOH	
CaCl ₂	

(10 Marks)

9. A new anti-rust product called NORUST has just come onto the market. The packaging reads: *Spray NORUST on your nails and dry for 12 hours above 40°C to form a protective anti-rust coat.* Caution: NORUST contains toxic particles.

Choice magazine has received a complaint about the product not working. **(4 marks)**

- a) Is NORUST a solid, liquid or gas? Explain your reasoning.
- b) What two safety recommendations would you add to the caution label on the packaging of NORUST?
10. Design an experiment to test whether the product works – and whether it is likely the customer didn't follow the instructions.

Include a labeled diagram of the set-up as part of the method.

(16 marks)